EDWARD KUBIAK

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Summary

Full stack developer, four years building and owning production React applications across Ohio's K-12 education ecosystem. Agents write most of my code now; that moved the work rather than removing it — the judgment, the verification and the consequences stay put, and building the tooling that enforces that is what my open-source work is about. Hired two weeks out of a full-stack certificate in August 2022 to replace an end-of-life AngularJS platform with React, I shipped that rewrite, opened it to districts in September 2024, and remain its sole architect. Creator of CAST v10.0.1, a local-first multi-agent control plane for Claude Code whose 42-table execution record is searchable, signed and predictive, shipped as 9 Homebrew packages backed by 3,323 tests; and of Compute Atlas (compute-atlas.com), an open, source-cited census of the U.S. grid-scale compute buildout — 1,413 facilities across 50 states, released as open data with a public API.

Skills

AI & Agent Systems: Claude Code (hooks, agents, skills), Claude API, CAST v10.0.1 (27 agents, 9 packages), Agent Orchestration & Dispatch, Hook Architecture, MCP (Model Context Protocol), OpenTelemetry (GenAI spans), Deterministic Agent-Reliability Tooling, RAG / Embeddings (FTS5, Ollama)
Languages & Frameworks: TypeScript, JavaScript (Node.js, ESM), Python, Rust, Bash, React 18/19, Next.js 16, Express 4/5, Flask, Vite, Tauri 2, TanStack Query v5 / Table v8, Radix UI, Tailwind CSS v4, Monaco Editor, xterm.js
Data: MS SQL Server, PostgreSQL, SQLite (better-sqlite3, FTS5), BigQuery, Drizzle ORM, SQLAlchemy, Server-Sent Events (SSE)
Testing & Delivery: Vitest, React Testing Library, BATS, pytest / unittest, GitHub Actions, Jenkins, Docker / Docker Compose, Traefik, Homebrew Packaging

AI-Assisted Engineering Practice

Condensed; full write-up and dated case studies at edwardkubiak.com/practice.

- Ceremony scales to blast radius, not line count: a one-line edit to an enforcement file gets a written plan; a doc typo gets applied inline.
- The prompt is the specification — if an agent would have to read four files before writing anything, that context belongs in the dispatch, with exact anchors.
- One unit per dispatch, sized to the agent's turn budget: an agent that exhausts it stops mid-sentence with no error, which reads exactly like success.
- Review time goes where it is cheap to be wrong — generated tests get read harder than implementations, because a wrong test is invisible.
- Verify mechanically, then read the verdict: confirm the change is on disk before judging whether it is good. A verdict is evidence about quality, never existence.
- Every new gate is mutation-tested — revert the fix, confirm the check goes red, restore. An assertion that never fails is indistinguishable from one that cannot.

Professional Experience

Applications Developer · META Solutions — Columbus, OH August 2022 — Present

React 18/19 · Vite 7 · TypeScript · TanStack Query v5 / Table v8 · Radix UI · Monaco Editor · Express 5 · Flask · MS SQL Server · PostgreSQL · Docker · Jenkins · Vitest

- Hired two weeks out of a full-stack certificate to rebuild CrossCheck — the EMIS data-validation platform Ohio districts use to catch state-reporting errors before they cost funding — replacing an end-of-life AngularJS 1.x + jQuery + PHP application. A ground-up rewrite across an architecture boundary, not a framework swap: built through year one, internal deployments September 2023, production January 2024, opened to districts September 2024. Sole architect and author of 1,706 of its 1,728 commits.
- Own it end to end: ~54,000 lines, 265 source files, 1,193 tests passing as of August 30, 2026. Led its migration off Create React App to Vite 7, off Bootstrap onto a test-guarded design-token system, and off component-level useEffect fetches onto a typed service and query layer — TanStack Query v5 and Table v8, Radix, Monaco for in-app SQL — one caller at a time, each with a colocated test. TypeScript conversion is incremental.
- Shipped security as artifacts, not assumptions: a CSP pinned by a test that fails when the policy changes, console and debugger stripping on production builds only, a DOMPurify link-hardening hook, and a lint rule failing the build if raw HTML injection appears outside one sanctioned module.
- Built SES-Wiki from an empty directory in March 2026 — React 19 + Vite 7, Express 5 on Node 20, strict TypeScript, ~21,300 lines, 340 tests: a storage adapter that fails loud on an unknown backend, ~25 REST endpoints with ETag revalidation and a typed error taxonomy mapped to status codes in one place, an append-only audit log on every

mutation, and the proxy-identity, bearer-validation and rate-limiting middleware. Multi-stage Docker image through Jenkins (lint, typecheck, tests — cheapest gate first) to a private registry and Portainer. In QA inside CrossCheck at same origin; cutover expected early 2027.

- Inherited and modernized the Customization Web Store (customizations.metasolutions.net), META's public PowerSchool catalog: Create React App to Vite 7, React 17 to 19, jQuery and Fancybox removed from a live page, an XSS vector closed with DOMPurify sanitization of API-sourced HTML, and an accessibility, SEO and dependency pass documented as seven dated findings. Also maintain the PowerSchool plugin customizations, including the Alert Builder notification system deployed across client districts.
- Own the E-Rate dashboard front end on a four-service Docker Compose platform (Flask + SQLAlchemy, PostgreSQL, a Python ingest scraper, Traefik, Jenkins) helping districts capture federal telecom discount funding. Ran an eight-dimension audit with adversarial verification in August 2026 — 98 findings, 3 critical, none refuted — then executed phase one: route guards on six authenticated routes reachable by direct navigation, a hardcoded dev-bypass login removed from the production bundle and grep-verified gone from the shipped artifact, npm audit 387 to 62, and a rebuilt applications table with debounced search, failed-edit rollback and a monotonic sequence race guard.
- Use Claude Code and CAST daily as production infrastructure — the tooling I publish as open source — with the review gate recorded rather than assumed: over the 30 days ending September 4, 2026, CrossCheck work passed 147 code-review verdicts, 14 returned with concerns and 2 blocked. It shows in git: 156 commits in all of 2025 against 519 in the first nine months of 2026, and a suite from 884 to 1,193 tests.

Open Source — AI Developer Tooling

Creator & Maintainer — CAST & Agent-Reliability Tools 2026 — Present

- Creator of CAST v10.0.1 — a local-first, open-source multi-agent control plane for Claude Code: 27 specialist agents with hook-driven dispatch and model-aware routing, hook-enforced quality gates, and a 42-table SQLite execution record that acts — full-text search over every session (cast ask), signed SHA-256 audit receipts (cast ledger --verify), telemetry-driven cost prediction (cast predict). v10 audited every gate against one question — what does this check report when the thing it guards did not happen? — and rebuilt the ones that reported success either way; new tests are mutation-tested against the bug they guard. 3,323 tests, zero cloud dependencies.
- Shipped it as 9 Homebrew packages plus an umbrella formula: the framework, cast-desktop (native Tauri 2 + React 19 + Rust with an embedded Express 5 + SQLite backend, 11 dashboard views and a PTY-backed terminal), the Claude Code Dashboard v3.0.0 (React 19 + TypeScript + Express 5 + SSE), and standalone packages for agent memory, health checks, journaling, MCP access, signed receipts and dispatch prediction.
- Built deterministic, zero-LLM agent-reliability tools, each shipped with CI — misfire and looptrip to PyPI and Homebrew, attest as a Homebrew formula and Claude Code plugin: misfire v0.2.0 audits CLAUDE.md adherence against real traces to find which rules agents actually ignore; attest v0.3.0 verifies a subagent's DONE claim against the real git delta; looptrip v0.1.2 trips duplicate-work loops at iteration 2 and reproduces prevented spend from a committed fixture.

Creator — Compute Atlas 2026 — Present

- Built and operate Compute Atlas (v1.32.0) — an open, source-cited census of the U.S. grid-scale compute buildout covering data centers, crypto-mining sites and the dedicated power generation contracted to feed them: 1,413 facilities across 50 states, 28.5 GW operational, 112.3 GW under construction and ~351.7 GW planned, every record carrying at least one public source and an explicit confidence level. Built solo on Next.js 16 · React 19 · TypeScript · MapLibre GL · Neon Postgres · Drizzle · Vercel.
- Shipped the whole product: an interactive MapLibre GL map with water and geology overlays, sortable tables, per-facility dossiers, reference hubs (rankings, power, operators, metros, states, AI-vs-crypto), a cited /learn tier and a CORS-open JSON API — the dataset released as open data (code MIT, data CC BY 4.0).
- Engineered an autonomous daily discovery pipeline: a bounded `claude -p` run that discovers, enriches and stages candidates behind a fail-closed kill switch, a heartbeat, extraction guards and a self-reverting cap, verifying every candidate source URL before staging, with human approval required before any record goes live. Extended the survey beyond capacity into its civic footprint — community opposition (litigation, moratoria, referendums, and the projects opposition defeated) and named stakeholders tied to specific sites under a mandatory-citation bar.

Earlier Career

Nine years of operations and people management preceded the career change — shift and crew leadership, training, performance management, client accounts.

Shift Supervisor · United Parcel Service — Columbus, OH September 2020 — August 2022

Account & Production Manager · Peabody Landscape Group — Columbus, OH March 2013 — June 2020

Education

- Full Stack Web Development Certificate — The Ohio State University · 2022
- Bachelor of Arts and Science in Geological Science — Ohio University · 2005–2009